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Approximate multi-objective optimization for integrated bus route design and service frequency setting

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ABSTRACT

This paper focuses on the transit network design and frequency setting problem for bus transit systems, where solutions are simultaneously generated in favor of the passengers and operator. While multi-objective solutions exist in the literature, simplifications and inconsistent assumptions make them hard to compare. For example; researchers have proposed solutions that report on a minimal fleet size without considering the service frequencies that guide the passenger's travel options, as a result, logical choices made by all passengers would cause the buses in the transit system to be overcrowded. To address this issue, we have developed a mathematical model which couples service frequencies and passenger assignment reflective of available choice options a passenger can take in the network. This approach can achieve lower operating costs while still ensuring that passengers can reach their destination at no detriment to journey times. We have constructed a complete set of multi-objective solutions, and compared them with those in the literature. The experiments demonstrate that our current solutions generally dominate those in the literature.

1. Introduction

Today, public transportation systems are a key contributor to community access to employment, education, retail, health, and social facilities in a metropolitan area. With increased development in major cities, if public transportation systems are not updated frequently, they become outdated, providing services that do not meet the current demands for travel. Indeed, many people opt to choose private vehicles over public transportation systems because of convenience, comfort, speed and practicality (Anable, 2005). To influence a modal shift, it is essential to promote other options such as improving the infrastructure of public transportation systems (Garling, 2005).

Reducing our reliance on private vehicles could solve a series of problems, including fossil energy consumption, air and noise pollution, traffic congestion, traffic accidents, and insufficient road usage (Özener and Özkan, 2020). However, numerous individuals will be hesitant to surrender the comfort, privacy and convenience of their own vehicles if other options are not viable. They are willing to accept traffic congestion, parking difficulties and high parking costs rather than use services they deem unreliable, slow and inconvenient. Therefore, to increase ridership on public transportation, a cost-effective, reliable and time-efficient transport system must be provided to the user (Banister, 2008).

Potential barriers of an efficient public transportation system include minimal waiting time, crowding, safety, bus driver friendliness, route directness, reliability and bus cleanliness (Stradling et al., 2007; dell'Olio et al., 2010). From these measures, waiting time, directness and reliability can all be improved through better network design. Bus routes are flexible to change their

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structure as opposed to fixed train lines. Ergo, when improving network design, the bus system should form the basis for public transportation options.

Clearly, a more effective transportation network that considers overall passenger satisfaction is critical in improving ridership. However, this must be balanced against the financial viability of the transport operators. Purely improving passenger satisfaction will come at a cost to the public transportation provider, as excessive services are not financially viable. Therefore, optimization methods can be applied to enhance public transportation systems to provide adequate service to the public while minimising cost to the operator. Through a compromising solution, a design can be implemented that is affordable to uphold and provide a level of service that can reduce dependency on private vehicles.

Therefore, this paper explores compromising solutions between the operator and passenger costs in a transit network design and frequency setting problem through the implementation of a multi-objective design. While there is significant literature regarding transit route design, simultaneous consideration of optimal routes and service frequencies has not been addressed with careful consideration of the frequency share rule (Baaj and Mahmassani, 1991). For example; the fleet size, based on service frequencies has been calculated under various simple assumptions that might cause capacity issues.

This paper proposes a multi-objective approach for designing transit routes and their frequencies that explicitly minimizes the costs for passengers and operator. This approach implements a graph-based method to compute all possible shortest paths, accurately calculating the shortest travel time along the proposed network route structure. A linear program is also used to maintain adequate frequencies for the proposed network, and a heuristic method is utilized to attempt to minimize the required fleet size. The model also considers the time passengers incur in waiting. It differs from the model proposed by Arbex and da Cunha (2015) because fleet size was not strictly minimized. Our algorithm follows a coupled frequency setting and passenger assignment algorithm that is able to minimize the total fleet size while ensuring that sufficient capacity is available for passengers. The waiting time calculation has also been often overlooked. Although modelled appropriately by Liang et al. (2020), the waiting estimations only account for the initial on-boarding and first transfer waiting time. If a passenger were to transfer two times, a passenger would encounter further waiting time than what was estimated.

The simulated annealing (SA) algorithm is wrapped around the objective function in a multi-objective framework. The algorithm seeks to obtain non-dominated solutions through the implementation of a simple probability function that attempts to improve by considering a candidate solution against the Pareto front. The probability function can select worse solutions by considering both objectives and measuring the trade-off between them by considering all dominant solutions in the Pareto frontier.

2. Literature review

2.1. Problem overview

The bus transit planning process includes five key stages which together describe all the decisions that need to be made for the network to operate. These stages are: route design, frequency setting, timetable development, bus and driver scheduling (Ewing and Cervero, 2010).

The first two stages are commonly referred to in the literature as the Transit Network Design Problem (TNDP) and the Transit Network Design Frequency Setting Problem (TNDFSP) respectively (Ceder and Wilson, 1986). TNDP concerns itself with designing routes, aiming to either minimize the average travel time of a passenger or reduce the total length of routes. Whereas, the TNDFSP includes the allocation of the frequency at which services are required to run, in an attempt to calculate the required fleet size. Therefore, the TNDFSP is an extension to the TNDP, where not only are the routes designed but the service frequencies set as well. The most optimal design requires solving all stages simultaneously. However, a considerable amount of computation is required to solve any stage (Kepaptsoglou and Karlaftis, 2009).

Decisions regarding timetable development and scheduling of buses and drivers can only occur once routes are designed and frequencies set. The overall cost of timetable development and scheduling components relies upon the foundations of the designed routes and service frequencies. Therefore, the design of routes and setting of service frequencies are more important than timetable development and scheduling of buses and drivers (Ceder and Wilson, 1986).

Traditionally, bus routes were constructed using seasoned transit planners' local knowledge, following simple guidelines and using ad hoc procedures (Fan and Machemehl, 2008). With these methods, networks usually evolve over time. New routes will be added gradually and the corresponding frequency will be updated to cope with the ever-changing demand (Arbex and da Cunha, 2015). Even though planners can obtain reasonable solutions, they are incapable of designing the whole network quickly. Using current methods, their efforts would likely prove insufficient for transforming a current network into a global optimum for large transit systems. If a complete restructure of the network were implemented, it would eliminate inefficiencies and inconsistencies in the network.

Studies have used computer-based optimization methods to evaluate the solutions in an attempt to outline problems with the existing networks (Bagloee and Ceder, 2011). Even with these methods, the search space is so large that optimal solutions will not likely be found. For any network configuration, there exists a vast set of potential routes. An entire network has to consider subsets of all these potential routes to observe the efficiency (removing unneeded segments). No individual route can be considered optimal without considering other routes in the network. Furthermore, complexities are added when considering the multi-objective nature of the problem. Other complexities include the problem's formulation, non-linearity and non-convexity, combinatorial complexity and NP-hardness (Baaj and Mahmassani, 1991; Chakroborty, 2003; Mauttone and Urquhart, 2009).

2.2. Objective functions

When designing the transit network and searching for a solution, many different objectives and metrics have been established. Proceeding from the work of Mauttone and Urquhart (2009) it has been a common effort to solve the TNDFSP considering multiple objectives. Common objectives within a multi-objective design include the minimisation of both operator and passenger costs. Other approaches, focus on a single objective; minimising the average travel time and the number of transfers respectively (Chakroborty, 2003; Szeto and Wu, 2011). These criteria focus on passenger satisfaction. Even though these two criteria are similar, minimising the number of transfers is not the same as minimising the total passenger travel time, as a bus route could potentially be long and serpentine away from the destination for a specific journey. A compromize between total passenger time and number of transfers has been analysed by Szeto and Jiang (2014), Arbex and da Cunha (2015), where it was suggested that strict minimisation of required transfers is more likely to satisfy the number of complaints from passengers. Even with the single objective approach, different solutions can be obtained by exploring various performance measures. Single objective minimisation designs in the literature include; waiting time (Newell, 1979), operational costs (Yan and Chen, 2002), total travel time (Yu et al., 2010) and directness (Khakbaz et al., 2017; Yu et al., 2012).

Other studies utilize a variety of objectives (Gao et al., 2004; Nikolić and Teodorović, 2014; Szeto and Wu, 2011; Amiripour et al., 2014; Jha et al., 2019). These objectives often compete against each other, but applying weights can balance the objectives. However, with the weighted approach, it is nontrivial to adjust the weights, as the two objectives may not be linear and may change the solution considerably. Also, it does not enable explicit consideration of all objectives, where the decision-maker would determine the appropriate solution. Furthermore, the selected weights of the objectives may not guarantee the decision-makers' preference and desired trade-off of the solution obtained. Hence, there has been a considerable amount of effort to solve the problem using non-dominated solutions (Zuo et al., 2015; Owais et al., 2016; Ruano-Daza et al., 2018; John, 2016). These solutions allow the decision-maker to select an appropriate solution for the network amongst a set of Pareto efficient solutions. By generating this Pareto set of solutions, a compromize between passenger and operating costs can be analysed. With different balancing of objectives and constraints in the literature, new solutions to the design problem can be explored.

Regarding the design of routes, the TNDP concerns itself with evaluation methods that follow a strict set of assumptions, such as passengers always using the shortest set of routes. The objective function is calculated, to a measurable degree, not considering the implications of stochastic events such as buses not arriving on time or passengers making less rational choices. Ruano-Daza et al. (2018) considered the implications of event-based modelling for a solution, by developing a post-processing stage that uses a simulation of discrete events as an evaluation measure. Another study that applied a simulation method in a similar way was by Yan et al. (2013), where simulated annealing was coupled with a Monte Carlo simulation to design networks with stochastic travel times.

2.3. Exact methods

Traditional optimization methods have difficulty solving this problem. Various attempts have been made to formulate the TNDP as a Mathematical Program (MP) to solve exactly (Baaj and Mahmassani, 1991, 1995; Ceder and Israeli, 1993). However, these methods faced difficulties when attempting to construct routes, struggling with solution representation. When route construction is used by researchers, it is found that even small-sized problem instances are out of reach for commercial branch-and-bound based optimizers. This was supported by Ceder and Israeli (1998), constructing routes formulated in an MP based framework, but was unable to produce efficient results. Cancela et al. (2015) supported that the problem was out of scope for commercial optimizers, by focusing purely on the mathematical formulation. Through the literature, it appears that only one study has focused on solving a non-fictitious network, representative of a small city (Borndörfer et al., 2007). Similar problems without service frequencies (such as the customized bus problem) have demonstrated that exact solution methods are able to solve the problem within practical time-frames (Huang et al., 2020). General reviews of the problem support that heuristics and metaheuristics are the best way to manage the complexity of the problem (Farahani et al., 2013; Kepaptsoglou and Karlaftis, 2009).

Mathematical programming is most useful if optimal solutions are found, and given the combinatorial nature of the problem, scalability is unpractical for larger networks. No MP approach exists in the frequency setting phase for the TNDFSP. However, there is potential for integrated attempts that combine exact methods with metaheuristic algorithms. The assignment of specific objectives and construction methods to assign decision variables can aid the solution process; demonstrated by Subramanian et al. (2012) when solving a vehicle routing problem. Limited integrated attempts have been explored that utilize both exact methods and metaheuristic algorithms (Iliopoulou and Kepaptsoglou, 2019; Walteros et al., 2015; Xu et al., 2014).

2.4. Approximate methods

The first heuristic approach, and notably the first mention of the TNDP was presented by Lampkin and Saalmans (1967). This involved developing skeleton routes by using shortest path and kth shortest path algorithms. Baaj and Mahmassani (1991), Mauttone and Urquhart (2009) have also explored this method, but due to the advancement in computation power, these methods have begun to fall out of favour, as the approach will continuously ask the designer to guide the algorithm. As the years progressed, researchers began to utilize metaheuristics. The most used approach across the literature are genetic algorithms (GA). However, until Owais and Osman (2018) the genetic algorithm has not been used to solve the entire TNDFSP. Where all routes were constructed from scratch, and the best route configuration chosen. Previous attempts using GAs typically involve performing crossovers or mutations

on existing route structures that are created prior to run of the algorithm (Ngamchai and Lovell, 2003; Fan and Machemehl, 2006a; Chien et al., 2001).

Regarding optimizing from an initial solution, (Pattnaik et al., 1998; Bielli et al., 2002) both used an external heuristic to generate an initial set of routes before solving the route and service frequency setting problem by using a GA. Chakroborty (2003) skipped over the route generation method altogether, using the existing structure of the network, and only then used a GA to update service frequencies. Tom and Mohan (2003) were the first to solve the route design and frequency setting simultaneously with a GA. Since then, attempts mostly solve the route design and frequency setting simultaneously, as this has been regarded as yielding more efficient solutions (Yu et al., 2011; Arbex and da Cunha, 2015; Cipriani et al., 2012; Walteros et al., 2015).

Simulated Annealing is less used, being explored by several studies (Fan and Machemehl, 2006b; Zhao, 2006; Yan et al., 2013; Ahmed et al., 2019). The most remarkable work utilising this method is explored by Ahmed et al. (2019), who tested different acceptance strategies to achieve the current state-of-the-art solutions. However, this was achieved by modified assumptions in the graph representation.

Tabu Search (TS) is very much overlooked, only discreetly being used by Fan and Machemehl (2008), Pacheco et al. (2009), Yao et al. (2014). This is because TS is prone to be trapped in a local-optima. However, as the size of the network increases, so does the solution space, even greedy algorithms show difficulty converging to local-optima. Attempts using TS also have not been explored on larger problem instances. Several attempts have been made that hybridize TS with other metaheuristics such as SA, to prevent premature convergence (Zhao et al., 2005). In a later study, this attempt was also used to handle network timetable development (Zhao and Zeng, 2007).

Recently, swarm intelligence (SI) has become of interest, due to GAs failing to cope with large networks. In the SI family, there exists Ant Colony Optimization, Particle Swarm Optimization (PSO) and Artificial Bee Colony (ABC). ACO mimics the behaviour activity of real-life ants, depositing pheromone trails such that the optimal path to food can be retained for the colony. Hu et al. (2005) was the first to develop this metaheuristic for network design. Later methods were improved upon by Blum and Mathew (2011), Szeto and Wu (2011), Bagloee and Ceder (2011), Yu et al. (2012), Yang et al. (2007). Similar to ACO, BCO mimics the foraging behaviour of bees. Szeto and Jiang (2014) developed this metaheuristic framework to design transit routes and set frequencies, whereas Nikolić and Teodorović (2014) developed BCO by modifying the service frequencies from a set of candidate routes. Similarly, An et al. (2020) optimized peak hour services to reduce crowding. PSO algorithms are less prevalent within network design, with one example provided by Kechagiopoulos and Beligiannis (2014). This is mainly due to PSO fundamentally using a continuous optimization framework. The structural design of the routes is inherently discrete, and as a result, this method was developed under a discrete framework. To improve real-time speed, many of these algorithms can benefit and are suited to parallel computing. However, only few studies have taken advantage of this (Agrawal and Mathew, 2004; Yu et al., 2010; Cipriani et al., 2012; Yang et al., 2007).

This paper outlines a new mathematical representation of the coupled frequency setting and passenger assignment problem to ensure that service frequencies satisfy demand while limiting required fleet size under given assumptions. Our approach calculates the fleet size alongside passenger waiting time whilst ensuring that buses are not overcrowded. Previous studies have not presented a concise and complete methodology for fleet size and waiting time calculations with proper validation of bus overcrowding constraints. In addition, all existing proposed solutions from the literature have been calculated under our methodology, allowing for a complete comparison of all solutions; validating fleet size and waiting time and promoting future research. Furthermore, a novel route generation heuristic has been developed to overcome connectivity issues for each stop in the road network. This heuristic guarantees feasible routes with simple swap and insert mechanisms that are common in meta-heuristic algorithms for vehicle-routing problems. All route network solution measures are calculated and bounded within a multi-objective formulation solved using SA. Multi-objective designs are common solved using genetic algorithms, but relatively unexplored using SA which exhibits significant convergence capabilities as well as being memory efficient (Ahmed et al., 2019).

3. Methodology

Before going into a formal definition of the problem, it is necessary to establish some terminology. A *stop* is defined as an individual location where passengers can board or disembark from a bus. Each stop forms a node in the *road network* which also consists of valid connections between stops and their associated travel costs (based on travel time). A *route* is an ordered set of stops connecting the stops to be visited by a bus. The set of all routes forms the *transit network*.

3.1. Problem statement

Given a set of stops and valid connections between them (the road network), determine a feasible and efficient set of bus routes (the transit network) and the required service frequencies, to satisfy the origin–destination (OD) demand between all pairs of stops. The objectives of the problem are to minimize both passenger and operator costs. It is assumed that buses are dedicated to a single route and cycle it in forward and reverse directions.

Inputs to the TNDFSP are consistent with previous studies, and include the following. Let S be the set of all stops on the road network, and C be the matrix of connection costs between every pair of stops, with elements $c_{i,j}$ representing the travel cost between stops i and j with $i, j \in S$. Any invalid connections can be penalized with arbitrarily large values in C. It is generally the case that C is symmetric for problems published in the literature, but symmetry is not a necessary requirement. D is the demand matrix for each origin–destination pair of stops, with $d_{i,j}$ giving a measure of demand between stops i and j, expressed as the number of passenger movements per unit time. Objective modifiers are required to specify the relative weights of particular terms in the objective function. These objective modifiers will be defined later.

3.2. Generating TNDP solutions

While TNDFSP is primarily a routing problem, the requirement to determine service frequencies on each route adds an additional layer of complexity, and an important aspect which has often lacked close attention in previous studies. In this study, a requirement is imposed that service frequencies must be sufficient to ensure that buses are not overloaded, under the assumption of a stationary demand profile. Applying this assumption does not make for a straightforward calculation of service frequencies, primarily because passengers may have multiple options to travel from their origin to destination. However, applying an additional assumption that passengers will make route choices which minimize their journey time (i.e. shortest path) provides a means to model passenger behaviour. The problem is challenging because there can be multiple shortest paths for passengers through a transit network potentially consisting of multiple transfers.

It is reasonable to assume that passengers will board the first bus that arrives at the current stop only if it provides the shortest path. Thus, it can be inferred that passengers for a given OD pair waiting at a particular stop will board compatible routes proportional to the relative service frequencies passing through that stop. This is referred as the *frequency share rule*, introduced by Baaj and Mahmassani (1991). Subsequent studies have applied this assumption (Liang et al., 2020; Arbex and da Cunha, 2015). We define the proportional distribution of demand for every OD pair across the multiple shortest paths as *passenger assignment*. Thus, service frequency and passenger assignment are tightly coupled. Changing the service frequencies will alter passenger assignment, which in turn may result in some buses being overloaded. For this reason, service frequency and passenger assignment must be calculated together, not separately. Later in this paper, it is shown that the equations governing the relationship between service frequency and passenger assignment are non-linear, and so a greedy algorithm is applied as described in the process below.

1. Obtain the set of routes constituting the transit network.
2. Calculate shortest paths for each OD pair.
3. Determine an initial proposed set of service frequencies for each route.
4. Calculate passenger assignment given relative service frequencies.
5. Calculate and apply universal multiplier α across all service frequencies to prevent overloading.
6. If α is close to 1 (within tolerance) stop.
7. Decrease the service frequency of underloaded routes to match peak loading, only if it improves the objective function. Go to 4.

An overarching metaheuristic algorithm described in Section 4, applies perturbations to the transit network structure. The frequency setting process is then applied for every candidate transit network generated during execution of the metaheuristic. The remainder of this Section 3 provides a formal description for steps 2–7, and we come back to approximate optimization of transit network structure in Section 4.

3.3. Shortest path calculation

The transit network is used to construct a directed graph, where transfers between routes are made possible whenever there are common stops between multiple routes. This graph representation then forms the basis for identifying plausible routes and connections between OD pairs.

We define a weighted graph $G = (V, E)$ where each stop $s \in S$ on route $k \in K$ is a unique vertex $v \in V$ in the graph G. Let $s(v)$ and $k(v)$ be respectively be the stop and route associated with vertex v, then we can use a tuple to define the vertex such that $v \equiv (k(v), s(v))$. The transit network is defined by an ordered set of stops for each route, in its nominal forward direction, namely, $R_k = \langle R_{k,1}, R_{k,2}, \ldots, R_{k,n_k} \rangle$ for $k \in K$ where $n_k = |R_k|$. We can then define the set of vertices as $V = \{v \equiv (k, s) : k \in K, s \in R_k\}$.

The edge set E includes connections of adjacent vertices to each other within each set R_k, in the direction of travel for a route. E also includes edges (v_1, v_2) representing passenger transfers (from one route to another), which connect any occurrences of identical stops within different route vertex sets (i.e. on different routes). Formally, $E = E_1 \cup E_1' \cup E_2$, where:

$$E_1 = \{(v_1, v_2) : v_1 \text{ is the direct predecessor of } v_2 \text{ in } R_k, k \in K\}$$
$$E_1' = \{(v_2, v_1) : (v_1, v_2) \in E_1\}$$
$$E_2 = \{(v_1, v_2) : s(v_1) = s(v_2); v_1 \in R_k, v_2 \in R_{k'}, k, k' \in K, k \neq k'\}$$

Every edge $e = (v_1, v_2) \in E$ has an associated cost c_e. For edges within a route $e \in E_1 \cup E_1'$, $c_e = c_{s(v_1),s(v_2)}$ the passenger trip cost between stops $s(v_1)$ and $s(v_2)$. For transfer edges $e \in E_2$, we set $c_e = M$, an arbitrarily large parameter value. This is to ensure that shortest paths for OD pairs will contain the minimum possible number of transfers.

Using this G, the shortest path cost and associated shortest paths for each origin destination pair can be determined. For this study, Floyd Warshall's shortest path algorithm (APSP) has been applied (Floyd, 1962). The returned cost of all paths that requires a transfer contains value M (multiple for each additional transfer). However, we post-process the returned costs from Floyd Warshall's algorithm to remove all additional costs of M and replace it with more suitable parameters α_1, α_2 and α_3. Where α_1, α_2 is the additional cost of encountering the first and second transfer and α_3 is the cost for encountering any additional transfers. This post-processing stage ensures that the large parameter value M is arbitrary, cannot be too large, and we will not overestimate transfer costs. It is sufficient to set M near to an estimate of the longest possible journey time cost, as this will ensure that no transfer edges will be used in the shortest path algorithm unless it is necessary.

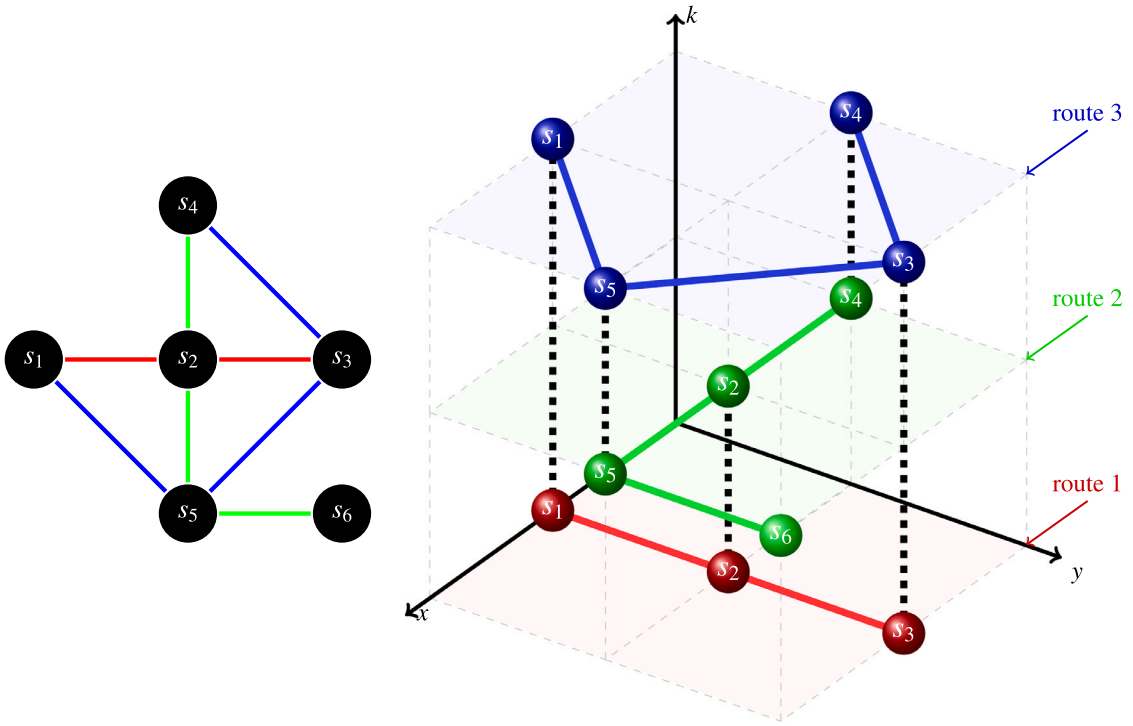


Fig. 1. Graph representation of a given transit network. Direct edges are depicted by straight lines and transfer edges by dashed lines

The shortest paths are stored within the sets Ψ_{od} for each OD pair $(o,d) \in OD$. Let $p = \langle p_1, p_2, \dots, p_{|p|} \rangle$ with all $p_i \in V$, be an ordered set representing a path within Ψ_{od}. Note that we will have $s(p_1) = o$ and $s(p_{|p|}) = d$. We denote the shortest path cost as $\delta_{od} = \delta_{od,1} + \delta_{od,2}$ where $\delta_{od,1}$ and $\delta_{od,2}$ are the respective totals of edge costs in E_1, E_1' and E_2 across shortest paths for (o,d). Note that $\delta_{od,1}$ represents the in-vehicle travel cost (i.e. travel time) for passengers, not including any waiting time for boarding or transfers.

An example of the graph representation for this problem is provided in Fig. 1 for a six stop road network serviced by three routes. Transfers are possible between routes at all stops except for stop s_6. The OD pair (s_1, s_6) is an example where the shortest path requires at least one transfer.

3.4. Initial proposed frequencies

For each OD pair, $(o,d) \in OD$, there may exist multiple shortest paths such that $|\Psi_{od}| \geq 1$. Because of this, the demand D_{od} is to be distributed between the paths $p \in \Psi_{od}$. For an initial passenger assignment, we base this distribution on the costs related to route length, with a higher assignment distribution on routes of the shortest length in terms of total travel time. The goal of this heuristic is to increase the service frequency on shorter routes, which is less costly than increasing the service frequency on longer routes, as it requires fewer buses per hour. On the contrary, (Arbex and da Cunha, 2015) assigns the passengers according to a logit normal distribution, such that passengers are randomly allocated. We do not adopt this methodology, as increasing service frequency on shorter routes will incline more passengers to be carried on the shorter routes (due to higher relative frequency). This method attempts to minimize the fleet size. Reducing fleet size directly corresponds to minimising the operational costs. The assignment procedure is described by Eqs. (1) and (2).

$$A_{od,p} = 1 - \frac{\sum_{k \in \Phi_{od,p}} l_k}{\sum_{q \in \Psi_{od}} \sum_{k \in \Phi_{od,q}} l_k} \qquad \forall od \in OD, p \in \Psi_{od} \tag{1}$$

$$X_{od,p} = \frac{d_{od} A_{od,p}}{\sum_{q \in \Psi_{od}} A_{od,q}} \qquad \forall od \in OD, p \in \Psi_{od} \tag{2}$$

Where, $A_{od,p}$ is a measure related to the proportion of passengers to assign to path p; $X_{od,p}$ is the number of passengers assigned to path $p \in \Psi_{od}$; l_k is the length of the route $k \in K$ based on C; $\Phi_{od,p}$ is the ordered set of routes used in path p.

Using the result from Eqs. (1) and (2), the initial frequency is set in Eq. (3). This equation aims to ensure the peak segment of each route will be filled to capacity. Since symmetric OD is not a necessary assumption, peak utilisation of each route must be taken

as the maximum of loading in the forward and reverse directions, indicated by $i = 0$ and $i = 1$ respectively in (3).

$$\hat{F}_k = \frac{1}{L} \max_{i\in\{0,1\}} \left(\max_{s\in R_k} \sum_{od\in OD} \sum_{p\in\Xi_{od,k,s,i}} X_{od,p} \right) \qquad \forall k \in K \tag{3}$$

Where, L is the maximum number of passengers allowed on a bus at any one point along a route $\Xi_{od,k,s,0}$ is the set of paths for origin–destination pair (o,d) using route k in its forward direction, passing through stop s, where $P_{od,k,s} = \{p \in \Psi_{od} : s \in \Theta_{od,p,k}, k \in \Phi_{od,p}\}$. $\Xi_{od,k,s,1}$ defined similar to $\Xi_{od,k,s}$, but for the reverse direction of route k. $\Theta_{od,p,k}$ is the set of stops passengers of (o,d) on path p pass through whilst on route $k \in \Phi_{od,p}$.

This initial set of service frequencies $\hat{F}_k$ requires further refinement and improvement. Since the passenger assignments $X_{od,p}$ in this step were generated by a heuristic, there is no guarantee that the relationship between passenger assignments and service frequency is in accordance with the frequency share assumption, that is, route choice is proportional to frequency (Baaj and Mahmassani, 1991). This is handled in the following subsection.

3.5. Passenger assignment and frequency rescaling

Using a set of proposed service frequencies, we apply the frequency share rule to determine a passenger assignment which is consistent with the relative service frequencies using Eq. (4).

$$X_{od,p} = \frac{d_{od}}{Q_{od,p}} \prod_{t\in\Gamma_{od}} \frac{\hat{F}_{|r(od,p,t)|}}{\sum_{k\in\Lambda_{od,p,t}} \hat{F}_k} \qquad \forall od \in OD, p \in \Psi_{od} \tag{4}$$

In Eq. (4) we define $\Gamma_{od} = \{1, \ldots, |\Phi_{od,p}|\}$ as the counting set through the number of routes used on every path in Ψ_{od}. The function $r(od,p,t)$ denotes a lookup which returns route $k \in K$ that is in position t of the ordered set $\Phi_{od,p}$, taking $+k$ and $-k$ if used in forward and reverse directions, respectively. $\Lambda_{od,p,t}$ is the set of routes that could be taken as alternatives to the route in position t of path p (inclusive of route $r(od,p,t)$). Formally, this can be written as:

$$\Lambda_{od,p,t} = \begin{cases} \cup_{q\in\Psi_{od}} \{r(od,q,1)\} & t = 1 \\ \cup_{q\in\Psi_{od}: r(od,p,\tau)=r(od,q,\tau), \tau\in\{1,\ldots,t-1\}} \{r(od,q,t)\} & t > 1 \end{cases}$$

Finally, $Q_{od,p}$ gives the number of paths that are using the same routes as p (inclusive of p), which can occur when routes share sequences of stops so there are multiple options as to where transfers occur. In the absence of any more specific information regarding the transit network, it is assumed that transfer locations are favoured equally by passengers in these situations. Specifically:

$$Q_{od,p} = |\{q \in \Psi_{od} : \Phi_{od,p} = \Phi_{od,q}\}|.$$

For illustration, consider an OD pair with three shortest paths which we label as $\Psi = \{1,2,3\}$, omitting od from subscript for brevity. Now assume the following routes are used in each path $\Phi_1 = \{1,3\}, \Phi_2 = \{2,3\}, \Phi_3 = \{1,4\}$. Note that we have $Q_p = 1$ for all p in this example. Let us assume that $d_{od} = 10$ and the following service frequencies for the four routes $F = [1,1,2,3]$.

$$X_1 = 10 \times \left(\frac{1}{1+1}\right) \times \left(\frac{2}{(2+3)}\right) = 2; X_2 = 10 \times \left(\frac{1}{1+1}\right) \times \left(\frac{2}{2}\right) = 5; X_1 = 10 \times \left(\frac{1}{1+1}\right) \times \left(\frac{3}{(2+3)}\right) = 3.$$

.

The main reason for calculating these passenger assignments $X_{od,p}$ is to determine whether the service frequencies provide sufficient capacity to avoid overloading. Specifically, it is required:

$$L \times F_k \geq \max_{i\in\{0,1\}} \left(\max_{s\in R_k} \sum_{od\in OD} \sum_{p\in\Xi_{od,k,s,i}} X_{od,p} \right) \quad \forall k \in K \tag{5}$$

In the event that Eq. (5) is violated, feasibility requires that the proposed service frequencies $\hat{F}_k$ be modified. Care must be taken in doing this, because the passenger assignments $X_{od,p}$ used to verify Eq. (5) were calculated from $\hat{F}_k$ values. Note that if every $\hat{F}_k$ was multiplied by the same constant α, that α would cancel out from top and bottom in Eq. (4). Thus, the service frequencies can be modified in the following way to ensure Eq. (5) is satisfied without causing any fluctuations in passenger assignment:

$$\alpha = \max\left(1, \max_{k\in K, i\in\{0,1\}, s\in R_k} \frac{\sum_{od\in OD} \sum_{p\in\Xi_{od,k,s,i}} X_{od,p}}{L \times F_k}\right) \tag{6}$$

$$F_k = \alpha \hat{F}_k \qquad \forall k \in K \tag{7}$$

Once we have a valid solution satisfying Eqs. (4) and (5), we can calculate the total number of transfers required by $\sum_{od} \sum_p (|\Phi_{od,p}| - 1) X_{od,p}$. In addition to calculating waiting time, for every required transfer there is a fixed penalty in time added, it is common to use 5 min (Baaj and Mahmassani, 1991).

3.6. Decrease frequency of underloaded routes

Given predefined routes, Eqs. (4) and (5) combined with a suitable objective function would effectively define a nonlinear mathematical program to minimize operator costs while accounting for the relationship between service frequency and patronage. Consider the objective function given in Eq. (8).

$$\min_{R,F} \quad 2\sum_{k\in K} l_k F_k \qquad \text{subject to (4), (5)} \tag{8}$$

Eq. (8) aims to minimize the total number of buses required on the network, a proxy for operator cost. This mathematical program is difficult to solve, but if the service frequencies are fixed $\hat{F}_k$, it is straight forward to calculate passenger assignment as we have shown. The addition of Eq. (7) is to ensure a feasible solution will always be obtained in this approach. This solution will not be guaranteed optimal for the nonlinear mathematical program, but applying a heuristic algorithm to iteratively perturb and improve the service frequencies will at least converge to a locally optimal solution.

Applying Eq. (7) to scale up all initial service frequencies to be feasible will cause some routes to be over supplied in terms of service frequency (i.e. some frequencies could be reduced and not cause capacity issues). Therefore, for uncrowded routes we calculate a new $\hat{F}_k$ and use Eqs. (4) and (7) again to recalculate F_k. Given the feasible values we have for F, updated values for $\hat{F}$ can be generated by tightening F using Eq. (3). This will then change the ratios between service frequencies and so the $X_{od,p}$ values will no longer adhere to the frequency share rule, and will violate Eq. (4). Thus, the process of applying Eqs. (4) and (7) must be repeated and terminating when no further improvement in the objective function (8) can be achieved. For further understanding of the methodology, a complete example is provided in Appendix B.

4. Multi-objective simulated annealing algorithm

This section explains the proposed multi-objective simulated annealing (MOSA) algorithm to solve the TNDFSP. The proposed algorithm utilizes a three phase search procedure. The first two phases focus purely on solving both the passenger and operator objectives independently following a traditional SA algorithm. The last phase is a multi-objective search, incorporating a mechanism that allows the search to move into regions that will improve upon the compromising solutions between both objectives found so far.

4.1. Multi-objective optimization problem specification

There are two competing objectives to be considered, passenger cost and operator costs. These are conflicting because passenger costs can be reduced by offering more direct connections in the routes and increasing the service frequency, but this results in higher operating costs. The following mathematical program provides a description of the problem that MOSA is applied to.

$$\min_{R,F} \quad f_1(R,F), f_2(R,F) \tag{9}$$

$$f_1(R,F) = \bar{T} + a_w \bar{W} + (a_1\pi_1 + a_2\pi_2 + a_3\pi_{>2}) \tag{10}$$

$$f_2(R,F) = 2\sum_{k\in K} l_k F_k \tag{11}$$

$$\text{subject to (4), (5)} \tag{12}$$

$$R \in \mathcal{R} \tag{13}$$

$$F_k \geq 0 \qquad \forall k \in K \tag{14}$$

Eq. (9) indicates that two objectives are to be minimized over decision variables relating to routes in the transit network R and service frequencies F. Eq. (10) defines the objective for passenger costs. Several yet to be defined parameters and variables are introduced here, and will be formally defined later in this section. Operator costs are reflected in the objective (11), noting that this is consistent with (8), and recalling that l_k is the length of route k which is itself a variable determined from R_k. Constraints (4) and (5) are applied in (12) to ensure the frequency share rule and bus capacity limit are both satisfied. In (13), we define $\mathcal{R}$ as the set of all feasible transit networks that fully connect all OD pairs, either directly or through one or more transfers. Finally, in (14) the domain for F is specified.

In Eq. (10) there are several terms involved in evaluating passenger cost. The first term $\bar{T}$ represents the average travel time (not including waiting time). It is calculated by Eq. (15), recalling that $\delta_{od,1}$ is the in-vehicle travel time component of the shortest paths for (o,d).

$$\bar{T} = \frac{1}{d_{TOT}} \sum_{od\in OD} d_{od}\delta_{od,1} \tag{15}$$

Note that $d_{TOT} = \sum_{od\in OD} d_{od}$. Average passenger waiting time is represented by $\bar{W}$, which is calculated by Eq. (16).

$$\bar{W} = \frac{1}{2d_{TOT}} \sum_{od\in OD} \sum_{p\in \Psi_{od}} \sum_{k\in \Phi_{od,p}} \frac{X_{od,p}}{F_k} \tag{16}$$

This is calculated based on the assumption that waiting times at boarding or transfers are uniformly distributed, and so will be half of the separation between consecutive buses on the route, which is half of $1/F_k$, an assumption supported by Arbex and da Cunha (2015), Jha et al. (2019). In the objective function, $\bar{W}$ is weighted by the coefficient a_w allowing the waiting time to be penalized more severely than in-vehicle travel time.

The last terms in (10) which are enclosed in parentheses, are to apply additional penalties to transfers. Variables π_1 and π_2 represent the proportion of passengers requiring one and two transfers respectively, with $\pi_{>2}$ for three or more transfers, and is calculated by Eq. (17).

$$\pi_j = \frac{1}{d_{TOT}} \sum_{od \in OD} d_{od} \mathbb{I}(\Psi_{od}, j) \tag{17}$$

The function $\mathbb{I}(\Gamma_{od}, j)$ is an indicator function returning 1 if there are j routes used in (o, d), and 0 otherwise, formally:

$$\mathbb{I}(A, n) = \begin{cases} 1, & |A| = n \\ 0, & |A| \neq n \end{cases}$$

Coefficients a_1, a_2 and a_3 allow different penalty weightings to be applied against one, two and three or more transfers respectively.

Including $\bar{T}, \pi_1, \pi_2$ and $\pi_{>2}$ in f_1 is consistent with many studies in the literature (Chakroborty, 2003). Including waiting time $\bar{W}$ in f_1, along with fleet size in f_2 is consistent with more recent studies (Arbex and da Cunha, 2015). Both objectives f_1 and f_2 are used to determine a set of non-dominated solutions for (9) to (14). An approach to achieving this is outlined in the remainder of this section.

4.2. Solution representation for MOSA

The solution representation used in any metaheuristic is crucial to its efficient and effective implementation. A number of different solution representations were explored in this research, and for the problem instances investigated (in which the road network did not directly connect every pair of nodes), a priority order based representation was found to be effective.

In the case of Mandl's 15 node network, there exists a total of 21 edge connections, and so it is not a fully connected network. A random allocation of stops forming a route is unlikely to be feasible, with no direct connection between some adjacent stops in the route. Rather than using explicit stop sequences for the solution representation, a priority order was found to be more effective. We formally define the priority order as the following:

Definition 4.1 (*Priority Order*)**.** The priority order is an ordered set of all stops in the network for an individual route. Route construction occurs based on this ordered set. The maximum length of the constructed route is defined by a control variable associated with the priority order, denoted as P_k which specifies the required stops within the route.

Each route has a unique priority order which is initialised randomly with the ordered set and P_k (bounded by minimum and maximum route size). From this design, feasible fully connected routes can be constructed by a simple heuristic.

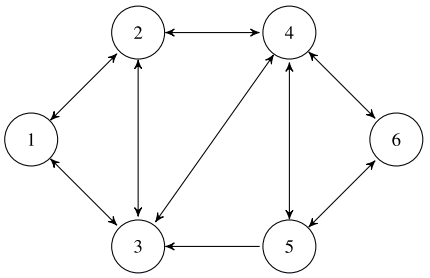
To construct a route with a given priority order, we begin with an empty route (no stops within the route). Stops are inserted into the last element of the route based on the priority order (only if connections are available). The first element in the priority order is always the first stop on the route. The next stop on the route is the element within the priority order that has the highest priority with an available connection to the first stop. This iterative process is repeated, constructing an expanding feasible route. A route will expand to a size that either has no more available connections to the last stop inserted on the route, or is fixed to the maximum route size defined by the variable P_k, a variable which is part of the priority order representation. P_k is bounded by control parameters such that $P_{\min} \leq P_k \leq P_{\max}$.

An example route generation phase can be observed in Fig. 2. This example produces a route observed in Fig. 2(c), where the available links are described in Fig. 2(a), and the priority order representation described in Fig. 2(b). Lastly, the route generation heuristic is shown graphically in Fig. 2(d).

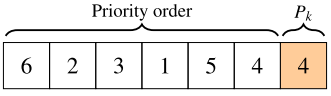
There are 4 iterations in the route construction heuristic as observed in Fig. 2(d) because $P_k = 4$. For example, if $P_k = 5$ there would be an additional iteration expanding route to include stop 1 (due to its priority and available connection). The first stop in the priority order will always be the first stop in the generated route; hence, stop 6 is the first element in Fig. 2(c) according to Fig. 2(b). The next stop is the first available connection in the priority order. Hence the second iteration in Fig. 2(d) generates stop 5, as it has the highest priority for the available connections to stop 6 as per Fig. 2(a). Priority does not affect on stop selection when no available links exist, as no connection is possible. The sequence of stops in the route is iteratively generated until there are no available connections from the last inserted stop in the route, or until the size of the route reaches the value of P_k.

This route construction method constructs feasible routes seamlessly. We can also perturb the priority order in any manner and still follow this procedure to construct a feasible route. This design is fundamental to how we perform operator moves within the metaheuristic to find efficient solutions. Perturbations on the priority order alter the routes, to generate neighbouring solutions.

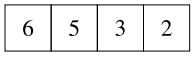
The entire network can be constructed using the priority order construction method on each route index $k \in K$. Even though all individual routes are viable, the collection of routes that form the transit network might exclude some nodes within the set. Hence, a repair method is introduced.



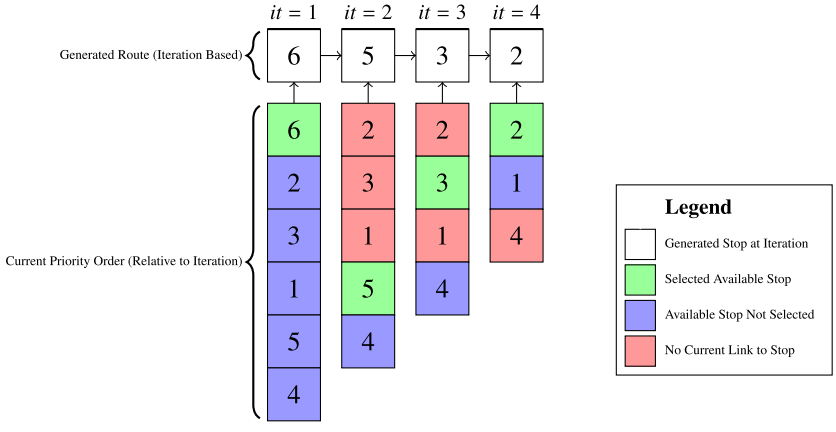
(a) Example network: available links from all stops.



(b) Example of the priority order representation for an individual route: priority order controls the generation of a route, P_k controls the length of the route.

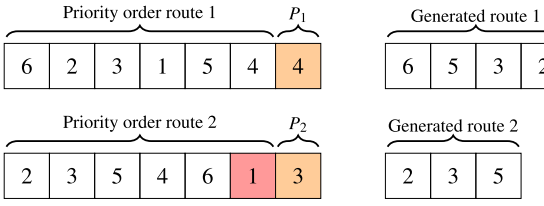


(c) Generated route via priority order representation

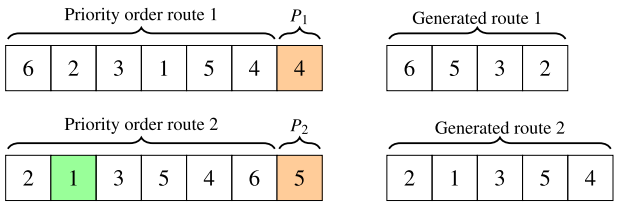


(d) Iteration based route generation from priority order and available links.

Fig. 2. Route generation.



(a) Infeasible 2 route network, missing stops 4 and 1. Stop 1 has been flagged for modification on route 2.



(b) Repairing route 2 by increasing the priority of stop 1 and increasing P_2.

Fig. 3. An example iteration of the repair routine for a network size of two routes. Iterations are repeated until the network becomes feasible. In this case the network is feasible in Figure Fig. 3(b).

The repair method finds a node that is not found in the route network set. Then the stop is moved up the priority order for the current shortest route in the network. Not only is the stop given higher priority on this selected route, but the route size variable is also increased to allow more stops to be visited. Essentially, the process modifies the shortest existing route in the network and forces the current missing stop to appear within a tour of the route. The route is reconstructed with the new priority order and increased stops within a route, and the missing nodes are checked again. If a node is still missing, the repair method is performed again and repeated until all nodes are present in the generated route network set. This routine is executed for every neighbouring solution generated within MOSA. The repair phase aims to restore feasibility to the solution, while avoiding significant disruption to the transit network structure. Fig. 3 illustrates this process.

Fig. 3 produces a network where all stops are present. Despite being repaired to a state where all stops are present, there is no guarantee that all OD pairs are connected within any number of transfers. This is penalised by the objective function.

Within MOSA, a number of perturbation operators are applied to the solution representation described above. Three perturbation operators have been defined for this purpose:

1. Randomly swap two stops in the priority order on the same route;
2. Randomly insert a stop into a new position on the same route;
3. Randomly adjust the max allowable route size in a given route.

Two control parameters are introduced to influence the frequency with which each of the perturbation operators are applied. These are denoted ξ_1, ξ_2 where random swaps are performed ξ_1 proportion of the time, insertions are done $1 - \xi_1$ proportion of the time, and route size perturbations are done ξ_2 proportion of the time in conjunction with either a swap or insertion.

4.3. Finding a pareto set using MOSA

Dominance is a term used to compare solution quality in multi-objective optimization. A solution u dominates u' if u is no worse than u' in all objectives, and solely better in at least one objective. Formally notated $u \prec u'$, if u dominates u'.

A solution u is said to be *Pareto efficient* if no other solution dominates it. We denote $\mathcal{P}$ to be all Pareto efficient solutions at the current state of all proposed solutions B. All solutions within $\mathcal{P}$ are said to be non-dominant to each other, as no solution can be observed as strictly better than another. Formally, the Pareto optimal solutions are defined in Eq. (18).

$$\mathcal{P} = \{u \in B : \nexists u' \in B \setminus \{u\}, f_1(u') < f_1(u) \wedge f_2(u') < f_2(u)\} \tag{18}$$

MOSA has three different phases, where each phase can be described as a separate independent search routine. Details on each phase are as follows:

Phase 1 SA search routine optimizing the passenger objective, storing multi-objective solutions into the set $\mathcal{P}$ if non-dominated.

Phase 2 SA search routine optimizing the operator objective, storing multi-objective solutions into the set $\mathcal{P}$ if non-dominated.

Phase 3 MOSA search routine, improve $\mathcal{P}$.

The first two phases adopt the same principles as a traditional simulated annealing algorithm (Kirkpatrick et al., 1983). The last phase is a dominance-based multi-objective simulated annealing algorithm (MOSA), that adopts the same methodology in Smith et al. (2008). The first two phases update $\mathcal{P}$ as new solutions are proposed, but no guidance is given to improve the overall Pareto efficient set. The last phase strictly attempts to improve on the efficiency of the solutions in $\mathcal{P}$. A mechanism that changes the acceptance criteria is utilized to exploit the search space to improve on the overall Pareto frontier. This procedure is detailed in Algorithm 2.

SA escapes local optima by accepting worse solutions with a probabilistic outcome. This occurs according to Eq. (19).

$$\Pr(\text{Acceptance}) = \exp(\frac{-\Delta E}{T}) \tag{19}$$

Where, ΔE represents the change in energy. We let $\Delta E = f(u') - f(u)$. This is the measurable difference between the candidate solution and the incumbent solution. T is a control variable within the algorithm representing the current temperature within the cooling schedule. A standard geometric cooling schedule is applied within the MOSA implementation, with the initial temperature T_0 determined following the approach of Johnson et al. (1989) such that a solutions should be accepted according to a desired ratio χ in the first temperature step.

In the first two phases of the MOSA, the measurable difference between u and u' are based on a single objective, $f_1(u)$ and $f_2(u)$ in phases 1 and 2 respectively. In phase three, the acceptance criterion is modified so that ΔE is determined with respect to the candidate set of known Pareto efficient solutions stored in $\mathcal{P}$. Using the current state of $\mathcal{P}$, we can observe how many individual solutions in $\mathcal{P}$ dominate u and u'. Any proposed solution u' that is dominated by less solutions from $\mathcal{P}$ than the current solution u will be automatically accepted.

Alternatively, when more solutions exist in $\mathcal{P}$ that dominate the candidate u' as opposed to the incumbent u, we utilize a weighted approach, where the weights are approximated using $\mathcal{P}$. Two weights are introduced, w_1, w_2 relating to the two objectives f_1 and f_2 respectively, such that they are more equally balanced according to the set of solutions $\mathcal{P}$, this is outlined in Eqs. (20) and (21).

$$w_1 = \frac{\sum_{x \in \mathcal{P}} f_2(x)}{\sum_{x \in \mathcal{P}} \left(f_1(x) + f_2(x)\right)} \tag{20}$$

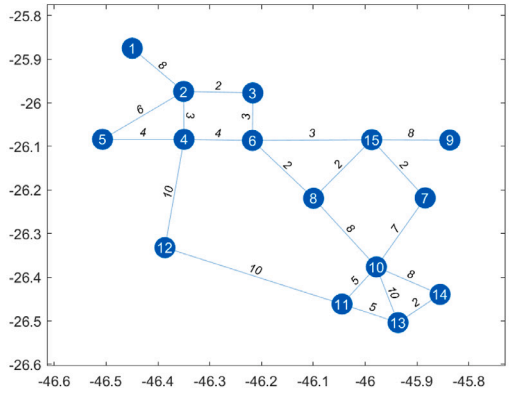
$$w_2 = \frac{\sum_{x \in \mathcal{P}} f_1(x)}{\sum_{x \in \mathcal{P}} \left(f_1(x) + f_2(x)\right)} \tag{21}$$

The weighting of w_1 and w_2 is updated for each new Pareto efficient solution found, as the current state of set $\mathcal{P}$ is modified. Therefore, the weight of w_1 and w_2 is updated accordingly. The measurable difference between u and u' is detailed in Eq. (22). Establishing the measurable difference allows the algorithm to exploit the search space by avoiding local-optima.

$$\Delta E = \max\left(0, w_1(f_1(u') - f_1(u)) + w_2(f_2(u') - f_2(u))\right) \tag{22}$$

Table 1
Penalty for transfer scenarios, proposed in Arbex and da Cunha (2015).

Scenario	1st transfer cost	2nd transfer cost
1	5	17
2	30	40
3	15	20



(a) Network topology.

O\D	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
0	0	400	200	60	80	150	75	75	30	160	30	25	35	0	0
1	400	0	50	120	20	180	90	90	15	130	20	10	10	5	0
2	200	50	0	40	60	180	90	90	15	45	20	10	10	5	0
3	60	120	40	0	50	100	50	50	15	240	40	25	10	5	0
4	80	20	60	50	0	50	25	25	10	120	20	15	5	0	0
5	150	180	180	100	50	0	100	100	30	880	60	15	15	10	0
6	75	90	90	50	25	100	0	50	15	440	35	10	10	5	0
7	75	90	90	50	25	100	50	0	15	440	35	10	10	5	0
8	30	15	15	15	10	30	15	15	0	140	20	5	0	0	0
9	160	130	45	240	120	880	440	440	140	0	600	250	500	200	0
10	30	20	20	40	20	60	35	35	20	600	0	75	95	15	0
11	25	10	10	25	15	15	10	10	5	250	75	0	70	0	0
12	35	10	10	10	5	15	10	10	0	500	95	70	0	45	0
13	0	5	5	5	0	10	5	5	0	200	15	0	45	0	0
14	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

(b) Demand for OD pairs

Fig. 4. Mandl's network.

Perturbation operators within MOSA only change the structure of the routes. The service frequencies of the network are assigned according to Eqs. (4) and (7). However, this frequency heuristic does not explore the possibility of reducing passenger waiting cost by increasing service frequencies. This is because Eqs. (4) and (7) and the associated frequency adjusting heuristic seek to determine tight service frequencies with a focus on minimising operator cost. To address this, at the end of the multi-objective search, we explore the optimized set $\mathcal{P}$ which contains the best network solutions with tightened service frequencies. We then adjust the service frequencies with a heuristic that incrementally reducing the parameter L which is the desired maximum peak loading of buses, and observe how solution quality changes in terms of f_1 and f_2, updating $\mathcal{P}$ based on the new solutions generated. This post-process heuristic is outlined in Algorithm 3.

5. Results

The proposed MOSA algorithm was tested on Mandl's network (Mandl, 1980), which comprises of 15 nodes and 21 bidirectional edge connection and is shown in Fig. 4. This network is representative of a group of cities in Switzerland. We have adopted the transfer scenarios, Table 1, utilised by Arbex and da Cunha (2015) to produce our results.

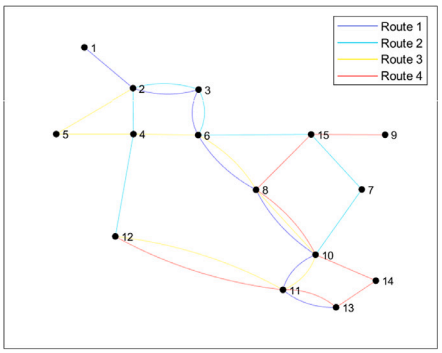
The following parameters were used over 3 different transfer scenarios:

- Capacity with loading factor: $L = 50$.
- Waiting time multiplier: $a_w = 2$.
- Number of nodes visited in a route: $P_{n\min} = 2, P_{n\max} = 8$.
- Minimum frequency of a route: $P_{f\min} = 1$
- Cost of more than 2 transfers or unreachable OD pair: $\alpha_3 = 100$
- Temperature calculated for $\chi = 0.5$ with 100 initial solutions.
- Steps before temperature reduction: 100
- Temperature reduction factor: 0.99
- Swap proportion $\xi_1 = 0.5$, route length proportion $\xi_2 = 0.2$.

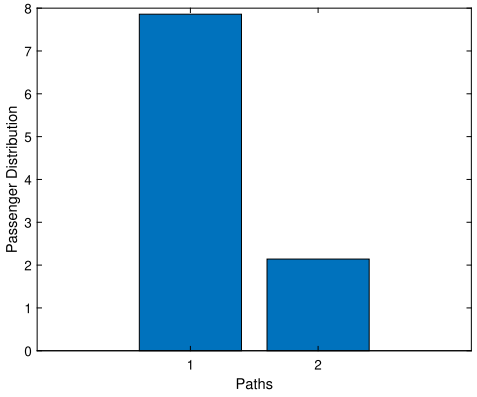
All algorithms were implemented in C++ and solved with 1 thread allocated on a SGI Altix XE Cluster. Each of the three phases of MOSA was given 3600 s of computation time.

5.1. Sample solution

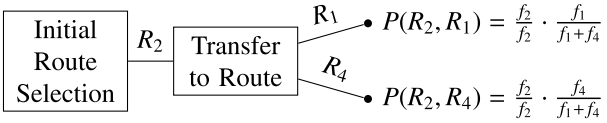
Figs. 5(b) and 5(c) demonstrates how passengers are assigned according to the network structure solution as seen in Fig. 5(a). One origin–destination pair was selected from the network structure to illustrate the passenger assignment scheme. Origin–destination



(a) Network route solution.



(b) Passenger distribution of two path options for origin-destination pair $(7, 13)$. The distribution is proportional to the service frequencies of the routes utilized at the transfer points, as both paths use the same initial route at the start of the journey.



(c) Probability of taking each shortest path for origin-destination journey 7 to 13 (only one choice for initial route selection).

Fig. 5. Passenger assignment.

pair $(7, 13)$ has two path options and requires a transfer. The two path options are shown in Fig. 5(c) and both paths start on route 2. Hence, the distribution of passengers is assigned relative to the frequency of the routes available when transferring. This is because passengers are equally likely to take both paths at the start of the journey because they utilize the same route, the deviation only occurs at what route they will transfer onto. The probability of a passenger taking a path is outlined in Fig. 5(c) and the assignment is shown in Fig. 5(b).

For the two paths, path option 1 assigns more passengers than option 2. This is because path option 1 transfers to a route that has a higher service frequency than option 2. This has been demonstrated in Fig. 5(c).

Fig. 6(a) shows the total number of passengers occupying capacity at each stop segment with respect to the route. Occupancy is what drives the frequency setting. Service frequencies are set so that there will always be capacity to accommodate the passengers according to the OD demand and subsequent passenger assignments. Fig. 6(b) shows the percentage capacity utilisation of each bus route at the given stop. The capacity utilisation of each bus route is shown to not exceed 100%, hence the routes, service frequencies and assignment of passengers produce a solution that is feasible.

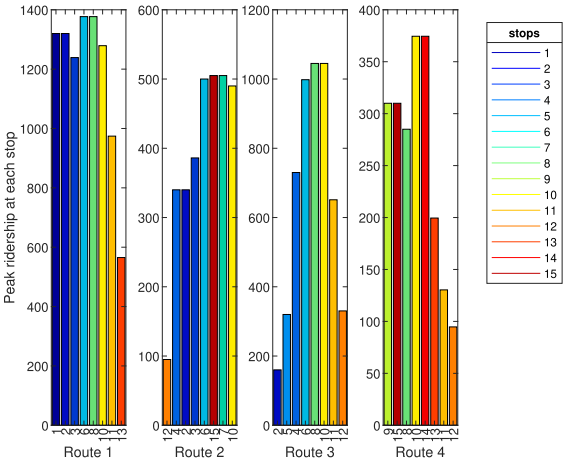
5.2. Solutions without consideration of waiting time

Results in this section demonstrate the trade-off between the costs associated with the user and the operator, when waiting time is not considered (i.e. set $(\bar{W}) = 0$ in (11)). In this case it is not possible to improve the average travel time of a passenger just by adjusting the service frequency for any given network. Therefore, all the solutions shown in the Pareto frontier in Fig. 7 are unique transit networks.

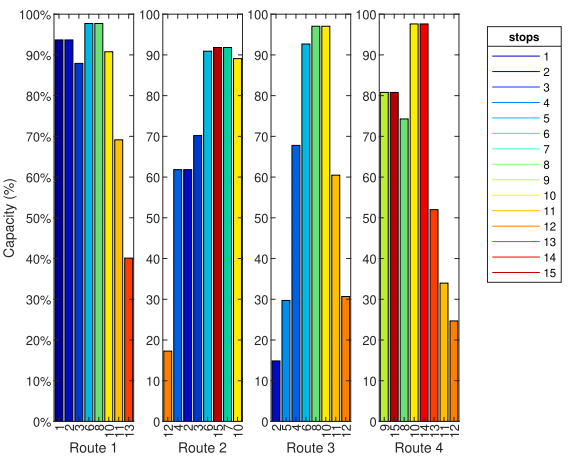
When the waiting time is not considered, the solution quality generally improves as the number of routes increases. Solutions with 12 routes are the most effective at minimising the average travel time, but this comes at a cost requiring a large fleet. When minimising fleet size, solutions with 11 routes were found to be the most effective. Solutions that balance both objectives effectively are networks with 8–10 routes. From the trends in the data, it is unlikely that networks with more than 12 routes will improve the Pareto front significantly, as the amount of improvement on the Pareto front becomes less measurable for each additional route.

5.3. Inclusion of waiting time

When the waiting time is considered, new Pareto efficient solutions can be found by altering the service frequencies of the routes. Increasing the services overall will lead to passengers waiting less time but will increase the overall fleet size. This is because the same network structure can appear multiple times in the Pareto frontier with the only difference being the frequency of the services. Adjusting the service frequencies to find these solutions is a computationally expensive effort. Hence, new service frequencies are only searched at the end of the MOSA procedure. Fig. 8 demonstrates how the post-processing frequency heuristic expands $\mathcal{P}$ by



(a) Patronage for each stop segment on the route when traversing in the forwards direction.



(b) Capacity percentage for each stop segment on the route when traversing in the backwards direction.

Fig. 6. Capacity levels for a solution

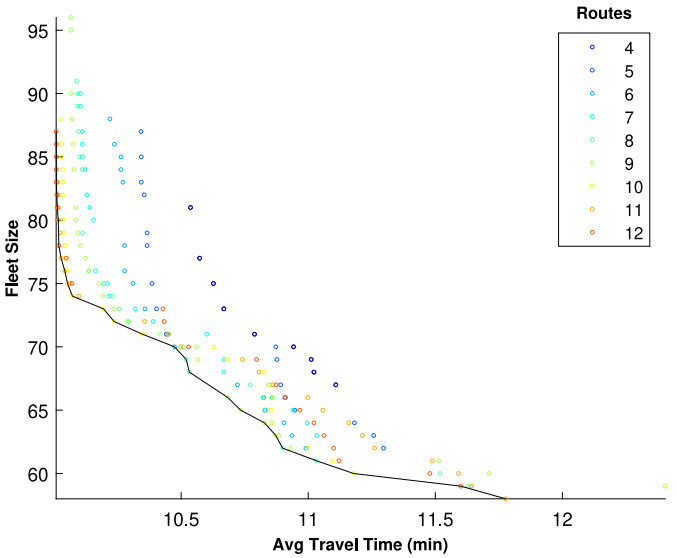


Fig. 7. Pareto frontier without waiting time cost.

altering the frequency of the solutions in $\mathcal{P}$. This post-processing stage is included throughout the remainder of our results and was included to demonstrate how the frequency setting problem in network design can lead to more compromising solutions.

Adjusting the service frequencies in favour of travel time results in a much more extensive Pareto front. Many of the modifications to the tightened solutions did not produce a new dominating solution within its range.

Fig. 9 compares the Pareto efficient solutions found by the MOSA with some recorded solutions in the literature. These solutions are available through (Ahern et al., 2020). The MOSA dominates the majority of the literature solutions with respect to the set route size.

In some of the literature solutions the fleet size is varied drastically across each route, in some cases only one bus is assigned to the route per hour. When this occurs, the average time spent waiting becomes heavily effected, as a passenger that is required to use a route with one service per hour is forced to wait a long amount of time. This is the main reason why some of the literature solutions appear to be of low quality. Furthermore, many of the literature solutions have calculated the required fleet size without ensuring that the service frequency can meet the demand based on the assignment of passengers through the frequency share rule. These solutions have been recorded but have been flagged as infeasible in Fig. 9.

In accordance with the frequency share rule applied in this study, many of the literature solutions are not feasible due to the service frequencies. More passengers take some routes as opposed to others, and thus some routes require the frequency to be higher

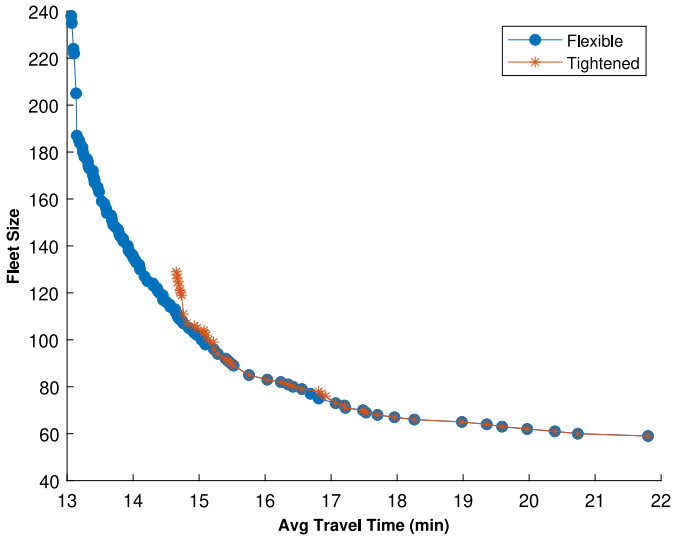
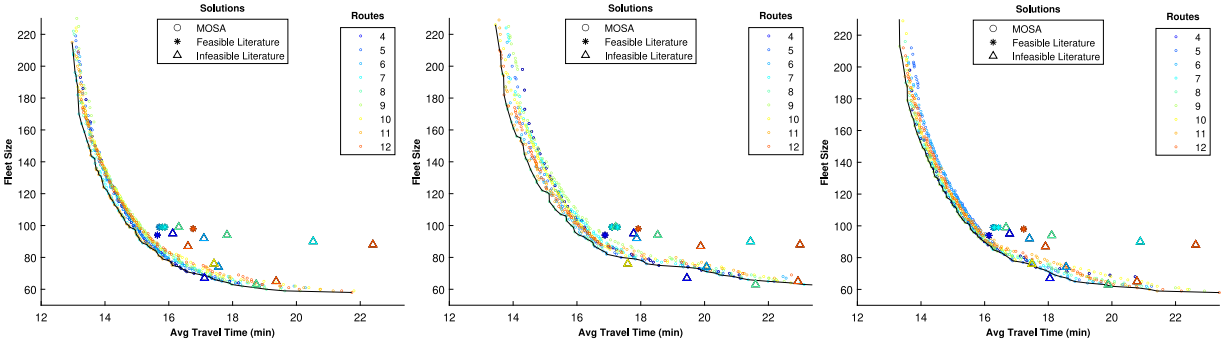
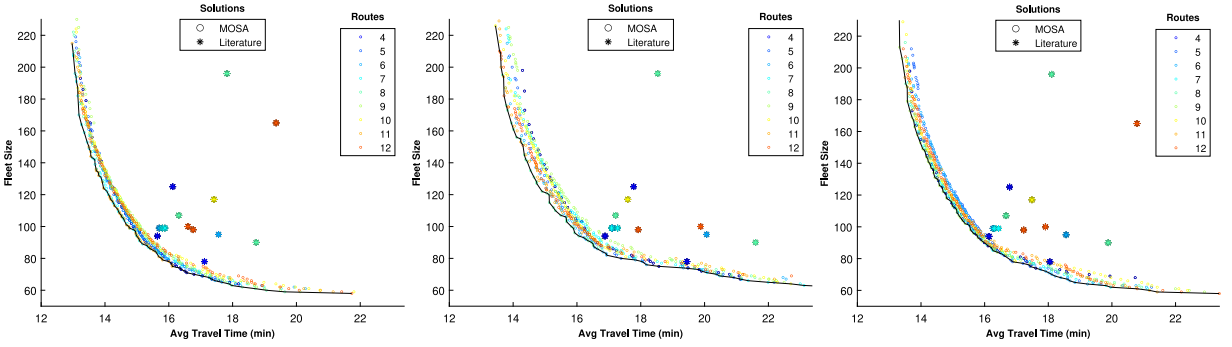


Fig. 8. Post-process frequency adjustment for a route network solutions consisting of 10 routes.



(a) Transfer 1 (b) Transfer 2 (c) Transfer 3

Fig. 9. MOSA vs literature solutions. Literature solutions as per recorded.

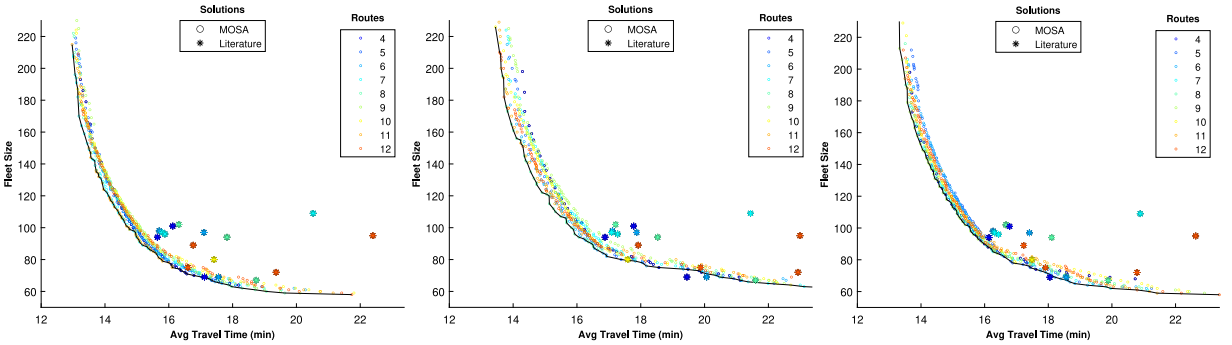


(a) Transfer 1 (b) Transfer 2 (c) Transfer 3

Fig. 10. Literature Solutions with service frequencies. Infeasible service frequencies are adjusted to feasibility.

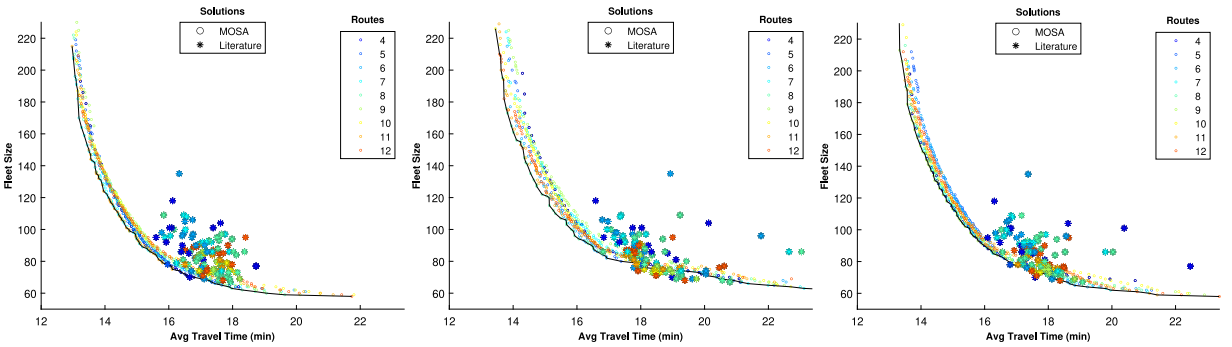
than what was reported when our methodology is applied. Fig. 10 takes the reported service frequencies of the literature solutions and corrects them for feasibility in accordance with Section 3.5. Note, this scales all frequencies by α as per Eq. (7).

However, many of the literature solutions are penalized by this adjustment. The scaling of the service frequencies to feasibility leaves many routes with wasted capacity. Fig. 11 repairs the service frequencies of the literature solutions as per Algorithm 4. Many



(a) Transfer 1 (b) Transfer 2 (c) Transfer 3

Fig. 11. MOSA vs Literature Solutions with service frequencies. Infeasible solutions based on service frequencies are adjusted to feasibility. Infeasible solutions are repaired using our heuristic method.



(a) Transfer 1 (b) Transfer 2 (c) Transfer 3

Fig. 12. MOSA vs literature network solutions without recorded service frequencies. Service frequencies are calculated with Algorithm 4.

of the solutions that were marked as infeasible have all been modified for feasibility. This is to preserve the fleet size specified by the published solutions while ensuring that the capacity limits are enforced by our methodology.

With the adjustments made to the published solutions, some solutions were found to be Pareto efficient. Nevertheless, many more of the solutions found within the MOSA dominate previously recorded solutions.

Many more solutions exist in the literature without considering the frequency setting problem or do not report on solutions with service frequencies. Fig. 12 compares the solutions found in the MOSA over three transfer scenarios against the literature's network solutions with service frequencies calculated using our heuristic method. The MOSA finds many more Pareto efficient solutions than the reported literature. Only some literature solutions are Pareto efficient under this methodology.

In regards to all results, the MOSA was shown to produce a set of solutions that generally captures the most optimal trade-off between the operator costs and passenger costs. However, some literature solutions were found to improve in some areas of our set of solutions. However, these instances were scarce as the majority of the non-dominating solutions were found by the MOSA.

5.4. Testing scalability to larger networks

To assess the scalability and generality of the proposed approaches, MOSA has been applied to generate a Pareto front for a published instances from Mumford (2013). These instances are referred to as Mumford0 (30 nodes, 90 links, 12 routes), Mumford1 (90 nodes, 210 links, 15 routes), Mumford2 (110 nodes, 385 links, 56 routes) and Mumford3 (127 nodes, 425 links, 60 routes). The first set of results demonstrates on Mumford0 how MOSA improves the Pareto front when given more time. In that experiment, the assumptions relating to passenger transfers have been modified to be consistent and comparable with previous studies which have investigated Mumford0. The second set of experiments applies the same assumptions underlying our Mandl results, and shows Pareto fronts for all four Mumford instances. Integrated optimisation of route design and frequency setting is computationally challenging and our experiments are consistent with this belief, however useful Pareto fronts could still be generated, albeit requiring lengthy CPU times.

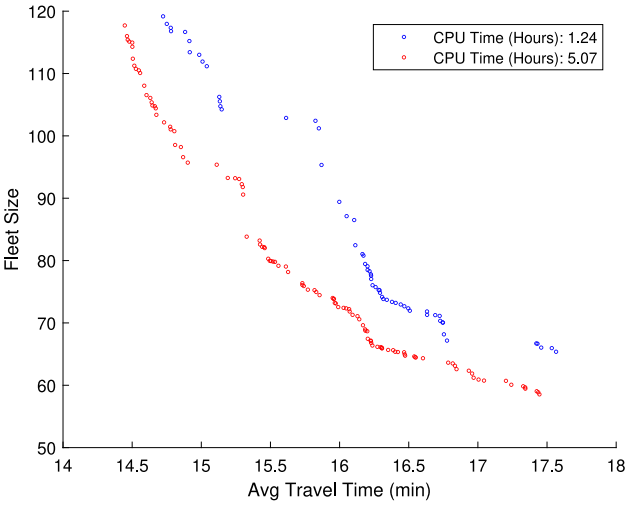


Fig. 13. Mumford0: 5 min transfer penalties, no waiting times.

5.4.1. Results with low transfer penalties and no waiting time

MOSA runs of varying length were performed on Mumford0 to determine CPU time required to settle on a reasonably stable Pareto front. It is important to note some difference in assumptions with previous studies (Mumford, 2013; Nayeem et al., 2014; John, 2016). Previously published literature for Mumford instances has not explicitly reported fleet size (aside from some very approximate calculations), and applied lesser penalties to passenger transfers. Based on the data available it was not possible to report on fleet size for a given solution, since the time-units of demand were not specified in the datasets. In order to effectively apply our methodology, it was assumed that demand was given as daily demand which then allows route frequency, passenger assignment and fleet size to be determined explicitly. Furthermore, to align with previous studies which have solved the Mumford instances, we have ignored the effects of waiting time by setting $\alpha_w = 0$, and have assumed 5 min transfer penalties (fixing $M = 5$). One consequence of these changes to our original assumptions is that passengers shortest-paths will not necessarily be those which require the minimum number of transfers, since transfers are not penalised so heavily.

The Pareto front for the Mumford0 problem instance under different run lengths can be observed in Fig. 13. Given the additional computational burden of integrated route design and frequency setting, significantly more CPU time was required in comparison to Mumford (2013).

Fig. 13 demonstrates how increasing the available CPU time significantly improved the Pareto front. However, compared to the earlier Mandl results, coverage of the Pareto front is less complete with lower solution density for solutions of around 15 min average travel time, and fleet size of around 90. The MOSA algorithm presented in this paper was not originally designed to work well with the modified assumptions in this experiment. Other experiments not presented here showed that for the larger Mumford instances, the MOSA algorithm tended to be attracted towards solutions with overly high number of passenger transfers. This is a result of the low transfer penalties applied in the path searching phase yielding passenger journeys (shortest paths) with many transfers, which are then later penalised by the objective function.

The best solution to minimise passenger average travel time found by MOSA was 14.47 min. It is worth noting that if were not for the work of Ahmed et al. (2019) reporting 14.09 min, the solution found by MOSA solution would have been state-of-the-art for a 12 route solution. Albeit, a significant amount of computational effort is needed to push the Pareto front towards this reported objective value.

5.4.2. Results with minimal transfers and waiting time

Enforcing the original assumptions (as per Mandl instance results) we produced Pareto frontiers for each Mumford problem instance as shown by Fig. 14. Varying levels of transfer penalty cost were investigated applying the three scenarios presented in Table 1. Mumford0 solutions tended to favour direct travel options, meaning that transfer scenarios showed little affect on passenger average travel time. The larger problem instances tended to require higher percentages of passenger transfers, and influence the solutions more noticeably. MOSA was able to produce reasonably complete Pareto fronts in Mumford0 and Mumford1 instances, but was less effective on the two larger instances. CPU times for these larger instances were in the order of 85 h. This suggests that further research is required to improve the scalability to larger instances of the integrated route design and frequency setting approach and MOSA algorithm presented in this study.

Overall, the fleet size metric proves to be more significant in the Mumford problem instances. This is because Pareto efficiency is not uniform from each solution. The results suggest that Pareto efficiency has marginal improvements as an individual objective is optimized purely. This means that utopia points are more easily visible (especially in Mumford1).

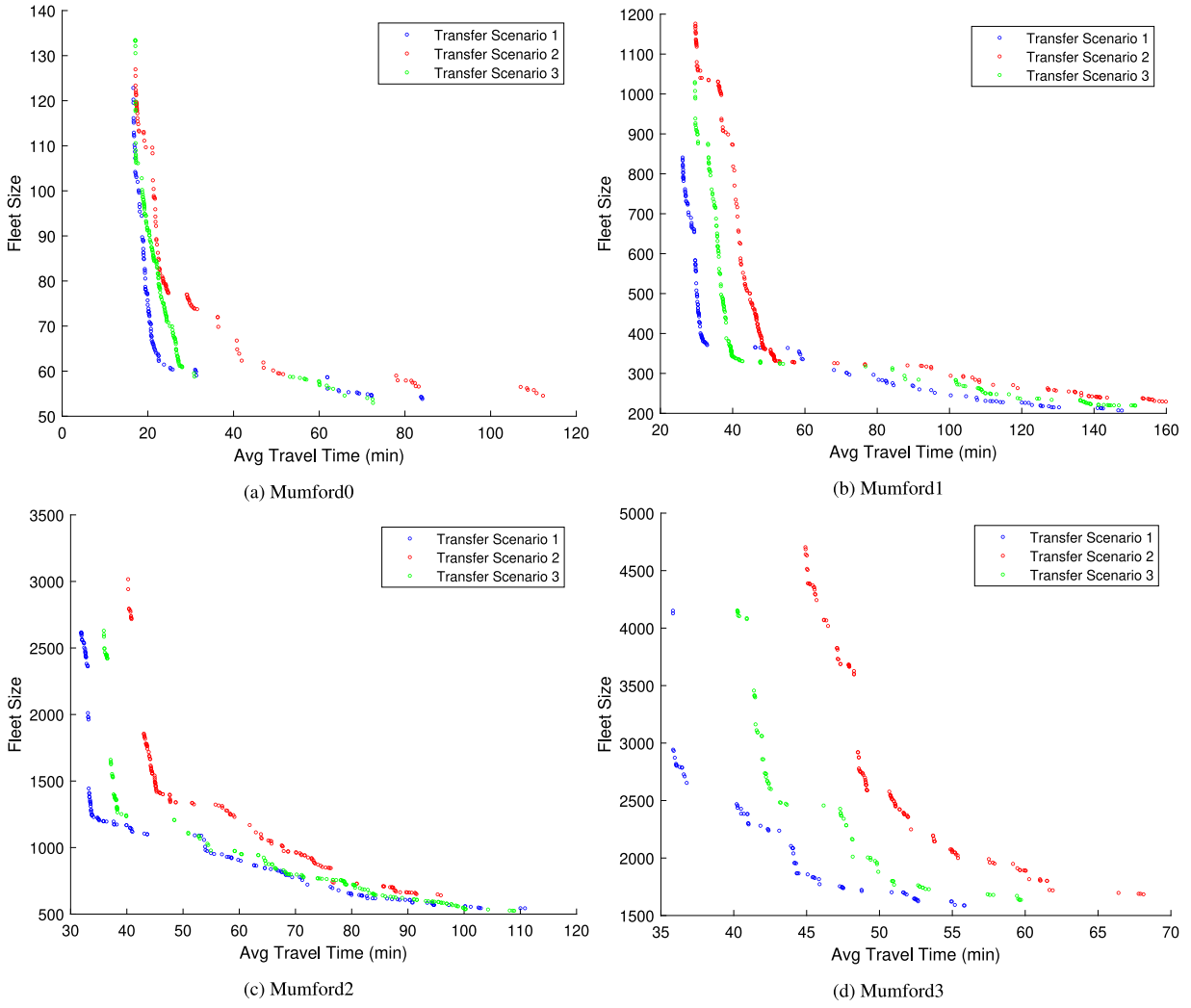


Fig. 14. Transfer Scenarios.

These results are available on the QUT data-finder.[1] It is worth noting that demand was always satisfied by our methodology.

6. Conclusion

An integrated solution method is proposed to simultaneously solve a transit network design and frequency setting problem. The solution approach includes a priority ordering route set construction method that aims to find a feasible neighbouring solution because the routes are constrained by the connections available at each stop. This approach allows accessibility for a range of operators to be used, previously difficult to implement due to the restrictions of available stop connections. Operators tested in this approach included swap, insert, and route size alteration.

Using a graph-based approach to calculate the average travel time of the route network considering all shortest paths for passengers, we constructed a mathematical programming model, solved through a heuristic decomposition to assign passengers according to a frequency share rule and ensure that service frequencies are tightened without exceeding capacity on routes. The main reason for this implementation is that by forcing the service frequencies to be as tight as possible, the fleet size is minimized, thereby approaching minimal operator costs. Operator and passenger costs are minimized simultaneously, which has not been addressed before while considering the frequency share rule.

A multi-objective simulated annealing algorithm was implemented to find the approximate Pareto front considering both objectives (i.e., minimising average travel time and fleet size). This metaheuristic was chosen because of its simplicity, low number

[1] https://researchdatafinder.qut.edu.au/display/n1527.

of hyper-parameters which are easy to fine tune, and easy extensions for multi-objective problems (Varadharajan and Rajendran, 2005; Xu and Qu, 2011; Amine, 2019). Experimental results show that the frequency assignment heuristic successfully ensures adequate service frequencies that meet the demand while limiting the number of buses required. Comparison was made against a large number of published solutions to a well-known problem in the literature. The proposed methodology was shown to generally produce Pareto dominant solutions, particularly when route capacity limits were strictly enforced. The multi-objective simulated annealing algorithm also produced an extensive Pareto front, proving its effectiveness to navigate through the search-space efficiently in terms of both objectives. Published solutions were also modified through our methodology to minimize fleet size and ensure feasibility, allowing for future researchers to make comparisons under a consistent framework.

Waiting time was also included in the model and was included in the operator costs of the network. Through inclusion, it was discovered that networks are less efficient when some routes operate at lower service frequencies. This is because these services provide some passengers with the shortest cost in terms of the journey spent in the vehicle, but forces the time spent waiting for a service to be large. In the Mandl problem instance, fleet size tends to be minimized more efficiently when the service frequencies are more diverse and have higher-level frequencies for shorter routes. Further research with other networks is required to determine the most efficient method to reduce average travel time for more general cases.

Considering fleet size in a multi-objective framework is a useful metric (particularly in the Pareto front) for the Mumford problem instances which historically have considered total route length costs as a kind of proxy for operator cost. The results observed in Ahmed et al. (2019) demonstrates that the total route length cost proxy does not provide adequate assistance for a decision-maker to select the most efficient solution, as all solutions have uniform separation in efficiency. The Pareto fronts we produced in the Mumford problem instance are more convex, and thus easier for a decision-maker to select a solutions that balances both objectives.

Future work will focus on improving the scalability of the proposed methods for application to larger and more complex urban transit environments. There is also an opportunity to explore the manipulation of service frequencies to reduce passenger waiting time during execution of the metaheuristic, as opposed to a post-processing step. Some metaheuristics that could enable to address this future work include the Global Best Harmony Search and its variants which have been used to address relevant large scale optimization problems (Cobos et al., 2020).

Another avenue for future work is to incorporate more realistic assumptions into the mathematical formulation of the frequency share rule. This study assumed all passengers choose the shortest transit path. Arbex and da Cunha (2015) considered the case where some passengers tend to make non-optimal decisions, which would be an interesting extension to the formulation proposed in this study.

6.1. Data

All Pareto efficient solutions specific to the transfer scenario are available on the QUT data-finder service.[2] Documentation in the QUT data-finder link explains the solution parameters in detail. Code is available online.[3]

CRediT authorship contribution statement

Zeke Ahern: Conceptualization, Methodology, Software, Formal analysis, Investigation, Resources, Data curation, Writing – original draft, Writing – review & editing, Visualization. **Alexander Paz:** Supervision, Writing – original draft, Writing – review & editing. **Paul Corry:** Supervision, Conceptualization, Methodology, Writing – original draft, Writing – review & editing.

Appendix A. Notation

See Table A.2.

Appendix B. Example

The transit network in Fig. B.15 with 6 stops and 4 routes is used to illustrate the passenger assignment model and frequency setting algorithm. Nominal forward direction of routes is indicated, noting that routes will run in both forward and reverse directions.

To avoid confusion and to aid in the simplicity of explanation, we will assume unidirectional travel on this network, with all OD pairs from higher to lower indexed nodes set to zero demand. For the remaining OD pairs, Table B.3 gives demand, the set of shortest paths Ψ_{od} and the associated route sequences $\Phi_{od,p}$ with travel in reverse direction (with respect to nominal direction) indicated as the negative route index. Vertices in each path from Ψ_{od} are coloured according to their associated routes.

For convenience, we will assume that initial frequencies for the four routes are given as $\hat{F} = (3, 1, 3, 1)$. For most $od \in OD$ and $p \in \Psi_{od}$ the passenger assignments $X_{od,p}$ are straight forward to calculate, so for illustration purposes we will focus on $od = (2, 5)$ which demonstrates how the approach handles more complex cases with multiple path options.

[2] https://researchdatafinder.qut.edu.au/display/n1527.

[3] https://gitfront.io/r/user-2155396/0d36a5b7ee124a4115e8f5f4fbba3483540935fe/ZA_Paper_Code/.

Table A.2
Key notation for network design and solution framework.

Notation	Description
Parameters	
S	Set of all stops on the network $S = \{1, 2, \dots, \lvert S \rvert\}$
$c_{i,j}$	Travel time/cost between all pairs of stops $i, j \in S$, with C denoting the full cost matrix
d_{od}	Demand given as number of passenger movements per unit time from between stops $o, d \in S$, with D denoting the full demand matrix, and we say that OD pair $od \in OD$
L	Desired maximum peak number of passengers on a bus
K	Set of routes on the transit network with $K = \{1, 2, \dots, \lvert K \rvert\}$, where $k \in K$ is chosen a priority
Decision variables	
R_k	Ordered set of stops visited for route k, and denote $R = \cup_{k \in K} R_k$
F_k	Frequency of route k in number of buses per unit time passing any given stop, and denote F as the vector of all frequencies $k \in K$
Related sets and variables	
l_k	Length of route k (i.e. sum of $c_{i,j}$ between consecutive pairs of stops on the route)
Ψ_{od}	Set of shortest paths discovered between OD pair
$\Phi_{od,p}$	Ordered set of routes used in path $p \in \Psi_{od}$
$\Xi_{od,k,s,i}$	Set of paths between OD pair using route k through stop s in forwards direction ($i = 0$) and reverse direction ($i = 1$)
$\delta_{od,1}$	Length of shortest paths between OD pair, not including transfer penalty costs
$\delta_{od,2}$	Length of shortest paths between OD pair, only including transfer penalty costs
$X_{od,p}$	Number of passengers (per unit time) between OD pair assigned to path $p \in \Psi_{od}$
$\bar{T}$	Average passenger travel time (not includiong transfer delays and waiting time)
$\bar{W}$	Average passenger waiting time including boarding and trafers
π_1, π_2	Proportion of passengers requiring 1 and 2 transfers respectively
$\pi_{>2}$	Proportion of passengers requiring 3 or more transfers
Other notation	
$f_1(R, F)$	Objective function relating to passenger cost, given routes R and frequencies F
$f_2(R, F)$	Objective function relating to operator cost, given routes R and frequencies F
a_w	Penalty coefficient for average waiting time in f_1
a_1, a_2, a_3	Penalty coefficients for π_1, π_2 and $\pi_{>2}$ respectively in f_1
MOSA control parameters	
χ	Desired ratio of accepted candidate solutions in the first temperature step
t_{step}	Number of iterations current temperature is retained
β	Cooling factor
$P_{n\max}$	Upper limit on the number of stops present within a service route
$P_{n\min}$	Lower limit on the number of stops present within a service route
$P_{f\min}$	Lower limit on the minimum frequency within a service route
ξ_1	Proportion of iterations a swap perturbation is applied, insertion applied $1 - \xi_1$ proportion of iterations
ξ_2	Proportion of iterations that a route length perturbation is applied

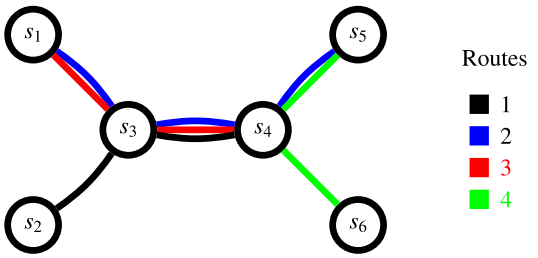


Fig. B.15. Example transit network.

Eq. (4) must now be applied to determine passenger assignments. The first two paths in $\Psi_{(2,5)}$ follow the same sequence of routes $\langle 1, 2 \rangle$, and so we have $Q_{(2,5),1} = Q_{(2,5),2} = 2$. Note that for the other two paths we have $Q_{(2,5),3} = Q_{(2,5),4} = 1$. Starting with the first path $\langle (1,2), (1,3), (2,3), (2,4), (2,5) \rangle$, the set of possible alternative routes at each stage t of this path is given by $\Lambda_{(2,5),1,t=1} = \{1, 3\}$ and $\Lambda_{(2,5),1,t=2} = \{2, 4\}$. Starting at stop 2, a passenger could have boarded on either route 1 or 3 ($\Lambda_{(2,5),1,t=1}$), and could then interchange onto either of route 2 or 4 ($\Lambda_{(2,5),1,t=2}$), noting that a different interchange stop would be required for route 4. This is consistent with the definition of $\Lambda_{od,p,t}$ given below Equation (4).

The data required to apply Equation (4) is now ready for the calculation:

$$X_{(2,5),1} = \frac{d_{(2,5)}}{Q_{(2,5),1}} \left(\frac{\hat{F}_1}{\hat{F}_1 + \hat{F}_3} \right) \left(\frac{\hat{F}_2}{\hat{F}_2 + \hat{F}_4} \right) = \frac{24}{2} \left(\frac{3}{3+3} \right) \left(\frac{1}{1+1} \right) = 3$$

Similarly, $X_{(2,5),2} = 3$.

This means that half of the passengers for $(2, 5)$ boarding route 1 will interchange at stop 3, and the other half will interchange at stop 4. This assumption is made in the absence of any specific information on passenger choice, but the calculation could be modified if required based on network specific information relating to interchange decisions.

Table B.3
Parameters.

od	d_{od}	Ψ_{od}	Φ_{od}
(1,2)	60	{⟨(3, 1), (3, 3), (3, 2)⟩}	{⟨3⟩}
(1,3)	40	{⟨(2, 1), (2, 3)⟩, ⟨(3, 1), (3, 3)⟩}	{⟨2⟩, ⟨3⟩}
(1,4)	50	{⟨(2, 1), (2, 3), (2, 4)⟩}	{⟨2⟩}
(1,5)	40	{⟨(2, 1), (2, 3), (2, 4), (2, 5)⟩}	{⟨2⟩}
(1,6)	32	{⟨(2, 1), (2, 3), (1, 3), (1, 4), (1, 6)⟩, ⟨(2, 1), (2, 3), (2, 4), (1, 4), (1, 6)⟩, ⟨(2, 1), (2, 3), (2, 4), (4, 4), (4, 6)⟩, ⟨(3, 1), (3, 3), (1, 3), (1, 4), (1, 6)⟩}	{⟨2, 1⟩, ⟨2, 1⟩, ⟨2, 4⟩, ⟨3, 1⟩}
(2,3)	64	{⟨(1, 2), (1, 3)⟩, ⟨(3, 2), (3, 3)⟩}	{⟨1⟩, ⟨−3⟩}
(2,4)	50	{⟨(1, 2), (1, 3), (1, 4)⟩}	{⟨1⟩}
(2,5)	24	{⟨(1, 2), (1, 3), (2, 3), (2, 4), (2, 5)⟩, ⟨(1, 2), (1, 3), (1, 4), (2, 4), (2, 5)⟩, ⟨(1, 2), (1, 3), (1, 4), (4, 4), (4, 5)⟩, ⟨(3, 2), (3, 3), (2, 3), (2, 4), (2, 5)⟩}	{⟨1, 2⟩, ⟨1, 2⟩, ⟨1, −4⟩, ⟨−3, 2⟩}
(2,6)	80	{⟨(1, 2), (1, 3), (1, 4), (1, 6)⟩}	{⟨1⟩}
(3,4)	48	{⟨(1, 3)(1, 4)⟩, ⟨(2, 3), (2, 4)⟩}	{⟨1⟩, ⟨2⟩}
(3,5)	30	{⟨(2, 3), (2, 4), (2, 5)⟩}	{⟨2⟩}
(3,6)	20	{⟨(1, 3), (1, 4), (1, 6)⟩}	{⟨1⟩}
(4,5)	30	{⟨(2, 4), (2, 5)⟩, ⟨(4, 4), (4, 5)⟩}	{⟨2⟩, ⟨−4⟩}
(4,6)	24	{⟨(1, 4), (1, 6)⟩, ⟨(4, 4), (4, 6)⟩}	{⟨1⟩, ⟨4⟩}
(5,6)	25	{⟨(4, 5), (4, 4), (4, 6)⟩}	{⟨4⟩}

Table B.4
Passenger assignments.

od	d_{od}	Φ_{od}	X_{od}
(1,2)	60	{⟨3⟩}	(60)
(1,3)	40	{⟨2⟩, ⟨3⟩}	(10,30)
(1,4)	50	{⟨2⟩}	(50)
(1,5)	40	{⟨2⟩}	(40)
(1,6)	32	{⟨2, 1⟩, ⟨2, 1⟩, ⟨2, 4⟩, ⟨3, 1⟩}	(3,3,2,24)
(2,3)	64	{⟨1⟩, ⟨−3⟩}	(32,32)
(2,4)	50	{⟨1⟩}	(50)
(2,5)	24	{⟨1, 2⟩, ⟨1, 2⟩, ⟨1, −4⟩, ⟨−3, 2⟩}	(3,3,6,12)
(2,6)	80	{⟨1⟩}	(80)
(3,4)	48	{⟨1⟩, ⟨2⟩}	(36,12)
(3,5)	30	{⟨2⟩}	(30)
(3,6)	20	{⟨1⟩}	(20)
(4,5)	30	{⟨2⟩, ⟨−4⟩}	(15,15)
(4,6)	24	{⟨1⟩, ⟨4⟩}	(18,6)
(5,6)	25	{⟨4⟩}	(25)

Table B.5
Frequency scaling and reduction.

Route	$\hat{F}_k$	Max load	Capacity	α_k	F_k	$\hat{F}_{k'}$
1	3	222	180	1.23	7.15	3.70
2	1	143	60	2.38	2.38	2.38
3	3	114	180	0.63	7.15	1.90
4	1	33	60	0.55	2.38	0.55
α				2.38		

Following this same process we can calculate passenger assignments for the other two paths:

$$X_{(2,5),3} = \frac{d_{(2,5)}}{Q_{(2,5),3}} \left(\frac{\hat{F}_1}{\hat{F}_1+\hat{F}_3}\right)\left(\frac{\hat{F}_4}{\hat{F}_2+\hat{F}_4}\right) = \frac{24}{1}\left(\frac{3}{3+3}\right)\left(\frac{1}{1+1}\right) = 6$$

$$X_{(2,5),4} = \frac{d_{(2,5)}}{Q_{(2,5),4}} \left(\frac{\hat{F}_3}{\hat{F}_1+\hat{F}_3}\right)\left(\frac{\hat{F}_2}{\hat{F}_2}\right) = \frac{24}{1}\left(\frac{3}{3+3}\right)\left(\frac{1}{1}\right) = 12$$

Intuitively this makes sense, with half of the passengers departing stop 2 choosing route 1 (paths 1,2 and 3), or route 3 (path 4), which both have the same frequency. There are three interchange choices from route 1, with half of route 1 passengers interchanging to route 2, and the other half to route 4 (again which have equal frequencies). Finally, of those passengers that interchange to route 2, as mentioned earlier half will make the interchange at stop 3 and the other half at stop 4. Table B.4 gives the calculated passenger assignments for all other OD pairs and paths.

Once we have the passenger assignments, the average loading of each route at each stop can be calculated, taking care to ensure direction of travel is taken into consideration as required in Eq. (6) (e.g. route 3 in $\Phi_{(2,3)}$). Assuming bus capacity $L = 60$, Table B.5 uses the ratio of calculated maximum load with route capacity as per Equation (6) to calculate a scaling factor α to uniformly scale route frequencies ensuring route capacities are not violated. This leads to the modified route frequencies in the second from right column. Finally, any routes with under-utilised capacity have their frequencies tightened to match passenger loading, through application of Eq. (3) shown in the far right column.

Table B.6
Convergence of passenger assignment, frequency scaling and reduction.

Iteration	1	2	3	4	5
$\hat{F}_1$	3	3.70	3.51	3.47	3.47
$\hat{F}_2$	1	2.38	2.54	2.57	2.57
$\hat{F}_3$	3	1.90	1.53	1.45	1.43
$\hat{F}_4$	1	0.55	0.61	0.63	0.64
α	2.38	1.11	1.04	1.01	1.00

Table B.7
Final frequencies and passenger assignment.

Route	$\hat{F}_k$	Max load	Capacity	α_k
1	3.47	207.86	207.94	1.00
2	2.57	154.37	154.30	1.00
3	1.43	85.77	86.01	1.00
4	0.64	38.42	38.35	1.00

After frequency reduction, the adjusted frequencies will alter the passenger assignment and possibly be infeasible. We therefore repeat the process that was just undertaken, but with the adjusted frequencies. Table B.6 shows that for this example the iterative process converges within two decimal places of α within five iterations.

Table B.7 shows the maximum loading for each route after the iterative process has converged. It can be seen that the route frequencies and capacities are very closely matched across all four routes. There is a slight capacity violation for routes 2 and 4, but it is within tolerance at less than 0.2% error.

This process was repeated with different initial frequencies: $(3, 1, 3, 1)$ shown above, $(1, 3, 1, 3)$, $(1, 10, 10, 1)$, $(10, 1, 1, 10)$ and $(1, 1, 1, 1)$. In all of these cases, the method converges to the same tightened frequencies $(3.47, 2, 57, 1.43, 0.64)$ in 4 to 6 iterations.

Appendix C. Algorithms

Algorithm 1 Simulated Annealing Algorithm with Pareto Efficient Solutions

Inputs:
$\mathcal{U}$ - Problem instance parameter dataset
f_Z - Objective function
$\mathcal{P}$ - Initial Pareto front of solutions
β - Temperature reduction factor
χ - Criterion probability
t_{step} - Iterations before temperature is reduced
$time_{max}$ - Time limit
Outputs:
$\mathcal{P}$ - Updated Pareto front of solutions
procedure SA
  $T_0 \leftarrow$ INITIAL_TEMP(χ) ▷ Sets temperature to acceptance criterian probability
  GETSOLUTIONS() ▷ Assign current, solution
  GETPRIORITYORDERING() ▷ Get priority order from best solution
  repeat
    for $it = 1, 2, .., t_{step}$ **do**
      $X_{can} =$ PERTURBATION_OPERATOR(X_{curr})
      UPDATEPARETO($\mathcal{P}, X_{can}$) ▷ Checks if pareto efficient solution is found and updates $\mathcal{P}$
      if $f_Z(X_{can}) < f_Z(X_{curr})$ **then** ▷ Accepts if candidate is more efficient
        $X_{curr} = X_{can}$
      else if $\exp\left(-\frac{f_Z(X_{can}) - f_Z(X_{curr})}{T}\right) \leq U(0, 1)$ **then** ▷ Acceptance criteria with objective function f_Z
        $X_{curr} = X_{can}$
      end if
    end for
    $T = \beta \cdot T$ ▷ Update temperature
  until $time_elapsed > time_{max}$ ▷ Time limit reached
  Return $\mathcal{P}$
end procedure

Algorithm 2 Multi-Objective Simulated Annealing

Inputs:
  $\mathcal{U}$ - Problem instance parameter dataset
  C - MOSA control parameter dataset
Outputs:
  $\mathcal{P}$ - Pareto front of solutions
procedure MOSA
  $\mathcal{P} = \text{SA}(\mathcal{U}, f_1(), \mathcal{P}, C)$ ▷ SA minimising passenger objective
  $\mathcal{P} = \mathcal{P} \cup \text{SA}(\mathcal{U}, f_2(), \mathcal{P}, C)$ ▷ SA minimising operator objective
  $\mathcal{P} = \mathcal{P} \cup \text{SA}(\mathcal{U}, \{f_1(), f_2()\}, \mathcal{P}, C)$ ▷ SA minimising P, with multi-objective acceptance criteria
  PostProcessFrequency($\mathcal{U}, \mathcal{P}$)
  Return $\mathcal{P}$
end procedure

Algorithm 3 Post Processing Frequencies from Pareto-frontier

Inputs:
  $\mathcal{U}$ - Problem instance parameter dataset
  $\mathcal{P}$ - Pareto front of solutions before frequencies are modified
Outputs:
  $\mathcal{P}$ - Pareto front of solutions with flexible frequencies
procedure PostProcessFrequency
  for $X = (R, F) \in \mathcal{P}$ **do**
    for $r = 1, 2, .., \lceil L/2 \rceil$ **do**
      Let $L' = L - r$, and let $\mathcal{U}' = (\mathcal{U} \setminus L) \cup \{L'\}$
      $\hat{F} =$ FrequencyCalculation($\mathcal{U}', R, F$)
      Let $\hat{X} = (R, \hat{F})$
      UpdatePareto($\mathcal{P}, \hat{X}$) ▷ Evaluate $f_1(\hat{X})$ and $f_2(\hat{X})$, check Pareto status
    end for
  end for
  Return $\mathcal{P}$
end procedure

Algorithm 4 Frequency Calculation

Inputs:
  $\mathcal{U}$ - Problem instance parameter dataset
  R - Transit network
  $\hat{F}$ - Initial route frequencies vector
Outputs:
  F - Locally optimal and feasible frequency vector
procedure FrequencyCalculation
  repeat
    $F =$FrequencyAdjustment($\hat{F}$) ▷ In accordance to Section 3.5
    $\hat{F} =$ SpareCapacity(F) ▷ Reduce frequency of under-loaded routes according to Section 3.6
  until $F \approx \hat{F}$ ▷ Once spare capacity is reduced sufficiently
  Return F
end procedure

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